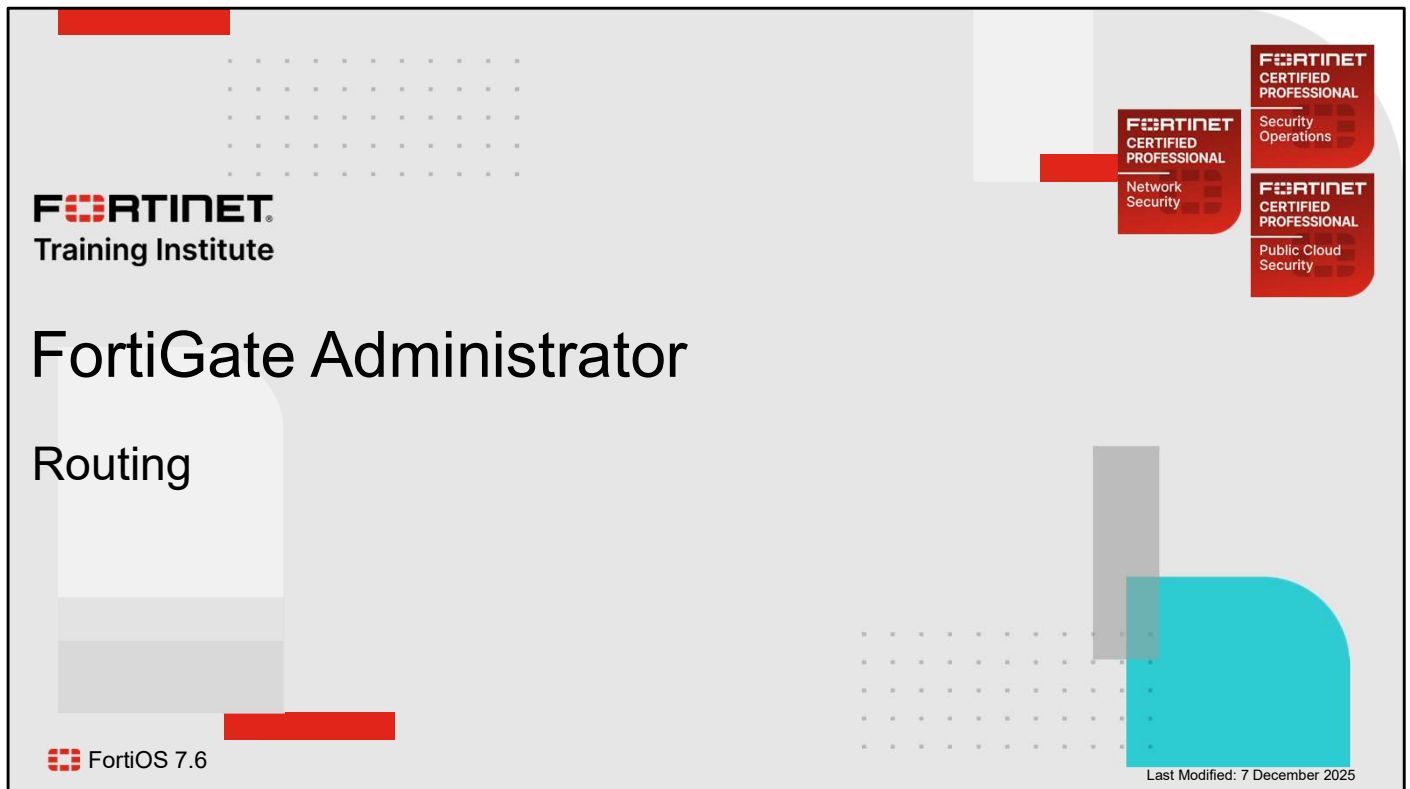
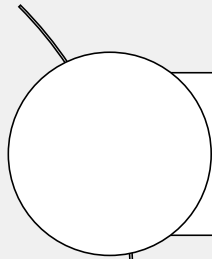


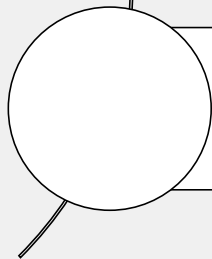


In this lesson, you will learn about the routing capabilities and features available on FortiGate.

Lesson Overview



Routing on FortiGate



ECMP Routing

In this lesson, you will learn about the topics shown on this slide.

Routing on FortiGate

Objectives

- Configure static routing
- Interpret the routing table on FortiGate

After completing this section, you should be able to achieve the objectives shown on this slide.

By demonstrating competence in routing on FortiGate, you should be able to implement static routing and understand the routing table.

What Is IP Routing?

- FortiGate acts as an IP router in network address translation (NAT) mode
 - Forwards packets between IP networks
 - Supports IPv4 and IPv6 routing
- IP routing:
 - Performed for firewall traffic and local-out traffic
 - Determines next hop (outgoing interface and gateway) for packet destination address
 - Next hop can be the destination router or another router along the path

When FortiGate operates in NAT mode—the default operation mode—FortiGate behaves as an IP router. An IP router is a device that forwards packets between IP networks. For that, a router performs IP routing, which is the process of determining the next hop to forward a packet to based on the packet destination IP address. FortiGate supports both IPv4 and IPv6 routing.

FortiGate performs routing for both firewall traffic (also known as user traffic) and local-out traffic. Firewall traffic is the traffic that travels through FortiGate. Local-out traffic is the traffic generated by FortiGate, usually for management purposes. For example, when you ping a device from FortiGate, that's local-out traffic. When FortiGate connects to FortiGuard to download the latest definitions, that's also local-out traffic.

What Is IP Routing? (Contd)

- Routing table:
 - Contains routes with next-hop information for a destination
 - Entries are checked during route lookup (best route selection)
 - *Best route*: most specific route to the destination
 - *Duplicate routes*: multiple routes to the same destination
 - Route attributes are used as tiebreakers for best route selection
- Routing precedes most security actions
 - Configure your security policies based on routing settings, not the opposite

Routers maintain a routing table. A routing table contains a series of entries, also known as routes. Each route in the routing table indicates the *next hop* for a particular destination. The next hop refers to the outgoing interface and gateway to use for forwarding the packet. The next hop can be the destination of the packet or another router along the path to the destination. If the next hop isn't the destination, the next router in the path routes the packet to the next hop. The routing process is repeated on each router along the path until the packet reaches its destination.

To route packets, FortiGate performs a route lookup to identify the best route to the destination. The best route is the most specific route to the destination. If FortiGate finds duplicate routes—multiple routes to the same destination—it uses various route attributes as a tiebreaker to determine the best route.

Routing takes place before most security features. For example, routing precedes firewall policy evaluation, content inspection, traffic shaping, and source NAT (SNAT). This means that the security actions that FortiGate performs depend on the outgoing interface determined by the routing process. This also means that your security policy configuration must follow your routing configuration, and not the opposite.

Route Lookup

- For any session, FortiGate performs a route lookup twice:
 - For the first packet sent by the originator
 - For the first reply packet coming from the responder
- Routing information is written to the session table
- All other packets for that session will use the same path
- No more route lookups done unless the session is impacted by a routing change
 - Route information on the session is flushed and new route lookups are performed

For each session, FortiGate performs two route lookups:

- For the first packet sent by the originator
- For the first reply packet coming from the responder

After completing these two lookups, FortiGate writes the routing information to its session table. Subsequent packets are routed according to the *session table*, not the routing table. So, all packets that belong to the same session follow the same path. However, there is an exception to this rule: if there is a change in the routing table that impacts the session, then FortiGate removes the route information for the session table, and then performs additional route lookups to rebuild this information.

RIB and FIB

- FortiGate maintains two tables containing routing information: RIB and FIB
- RIB
 - Standard routing table containing active (or best) connected, static, and dynamic routes
 - Visible from the GUI and CLI
- FIB
 - Routing table from a kernel perspective
 - Composed mostly of RIB entries, plus some system-specific entries
 - Used for route lookups
 - Visible from the CLI with the command `get router info kernel`

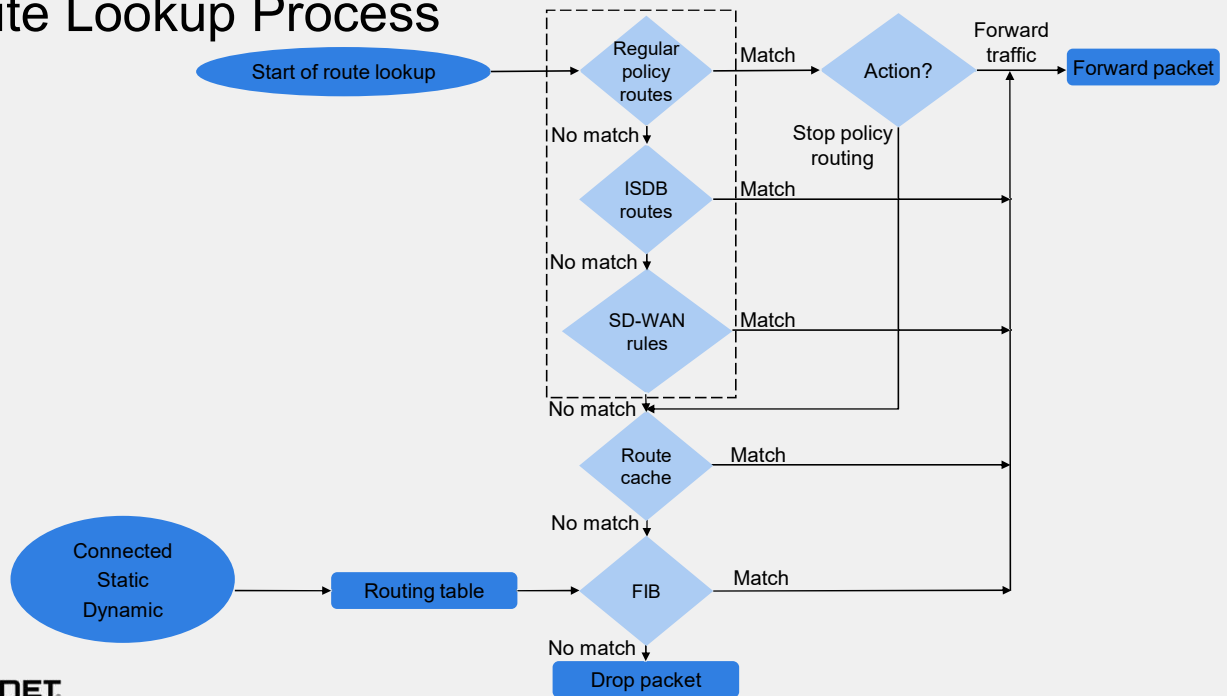
FortiGate maintains its routing information in two tables: RIB and FIB. The routing table, also known as the routing information base (RIB), is a standard routing table containing active (or the best) connected, static, and dynamic routes. The forwarding information base (FIB) can be described as the routing table from the kernel point of view, and contains mostly RIB entries plus some system-specific entries required by FortiOS.

When FortiGate performs a route lookup, it checks the FIB and not the RIB. However, because the FIB is composed mostly of RIB entries, a route lookup mainly involves checking routes from the RIB. For this reason, the route lookup is often referred to as the routing table lookup process. Nonetheless, it is more accurate to refer to a route lookup as a FIB lookup process.

You can display the RIB entries on the FortiGate GUI and CLI. You can display FIB entries on the FortiGate CLI only.

This lesson focuses on the RIB (or routing table) only. You will learn more about it, including how to monitor its entries, in this lesson.

Route Lookup Process



The flowchart on this slide describes the route lookup process in FortiOS.

First, FortiGate checks the policy routes. If there is a match, and the selected action is **Forward Traffic**, FortiGate routes the packet accordingly, as long as the policy route passes the forwarding information base (FIB) validation process. If the selected action is **Stop Policy Routing**, FortiGate moves on to check its route cache, if applicable.

If the packet doesn't match any of the policy routes, FortiGate moves on to check the ISDB routes first, and then the SD-WAN rules.

Next, FortiGate checks the FIB, which is the table that is used to perform standard routing. If the packet doesn't match any of the routes in the FIB, FortiGate drops the packet and sends an ICMP destination network unreachable message to the sender.

Static Routes

- Configured *manually*, by an administrator
- Simple matching of packets to a route, based on the packet destination IP address

Network > Static Routes

Edit Static Route

Destination ⓘ	Subnet Internet Service
	0.0.0.0/0.0.0.0
Gateway Address	10.200.1.254
Interface	port1
Administrative Distance ⓘ	10
Comments	Write a comment... 0/255
Status	Enabled Disabled
Advanced Options	
Priority ⓘ	1

Default route

One type of manually configured route is called a static route. When you configure a static route, you are telling FortiGate, “When you see a packet whose destination is within a specific range, send it through a specific network interface, towards a specific router.” You can also configure the distance and priority so that FortiGate can identify the best route to any destination matching multiple routes. You will learn about distance and priority in this lesson.

For example, in simple home networks, DHCP automatically retrieves and configures a route. Your modem then sends all outgoing traffic through your ISP internet router, which can relay packets to their destination. This is typically referred to as a default route, because all traffic not matching any other routes will, by default, be routed using this route. The example shown on this slide is a default route. The destination subnet value of 0.0.0.0/0.0.0.0 matches all addresses within any subnet. Most FortiGate devices deployed at the edge of the network have at least one of these default routes to ensure internet traffic is forwarded to the ISP network.

Static routes are not needed for subnets to which FortiGate has direct Layer 2 connectivity.

Static Routes With Named Addresses

- Firewall addresses set to type **Subnet** or **FQDN** can be used as destinations for static routes

Policy & Objects > Addresses

New Address

Name	Remote_Subnets
Color	Change
Interface	☐ any
Type	Subnet
IP/Netmask	Subnet
Routing configuration	☒
Comments	

IP Range
FQDN
Geography
Dynamic
Device (MAC Address)

Network > Static Routes

New Static Route

Destination	Subnet Named Address Internet Service
	Remote_Subnets
Gateway Address	100.64.2.254
Interface	port2
Administrative Distance	10
Comments	Write a comment... 0/255
Status	☒ Enabled ☐ Disabled

Advanced Options

If you create a firewall address object with the type **Subnet** or **FQDN**, you can use that firewall address as the destination of one or more static routes. First, enable **Routing configuration** in the firewall address configuration. After you enable it, the firewall address object becomes available for use in the **Destination** drop-down list for static routes with named addresses.

Internet Services Routing

- Route well-known internet services through specific interfaces

Policy & Objects > Internet Service Database

+ Create new

IP Address Lookup

Search

☐	Name	Direction	Number of Entries	Ref.
☐	Amazon-AWS	Both	6,596	0
☐	Amazon-AWS.API.Gateway	Both	139	0
☐	Amazon-AWS.AppFlow	Both	35	0
☐	Amazon-AWS.Chime.Meetings	Both	41	0
☐	Amazon-AWS.Chime.Voice.Connector	Both	22	0

Database containing IP addresses, protocols, and port numbers used by most common Internet services

Network > Static Routes

New Static Route

Destination: Subnet Named Address Internet Service

Gateway Address: 10.200.1.254

Interface: port1

Comments: Write a comment... 0/255

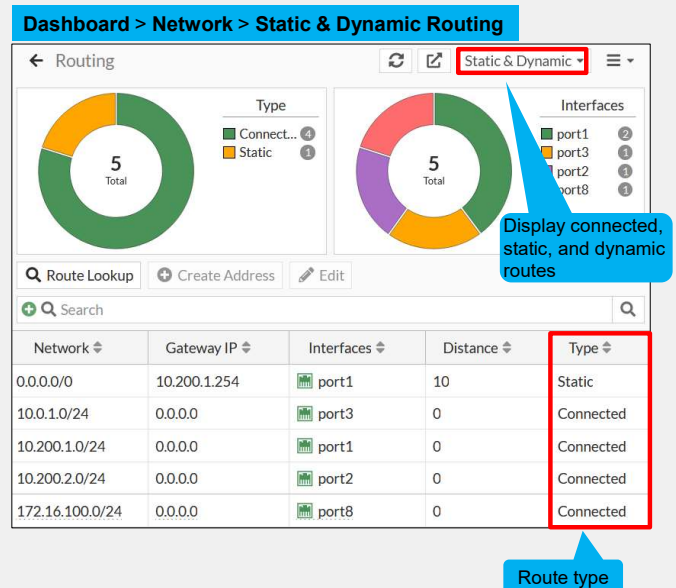
Status: Enabled Disabled

What happens if you need to route traffic to a public internet service (such as Amazon-AWS or Apple Store) through a specific WAN link? Say you have two ISPs and you want to route Netflix traffic through one ISP and all your other internet traffic through the other ISP. To achieve this goal, you need to know the Netflix IP addresses and configure the static route. After that, you must frequently check that none of the IP addresses have changed. The internet service database (ISDB) helps make this type of routing easier and simpler. ISDB entries are applied to static routes to selectively route traffic through specific WAN interfaces.

Even though they are configured as static routes, ISDB routes are actually policy routes and take precedence over any other routes in the routing table. As such, ISDB routes are added to the policy routing table.

Routing Monitor

- Routing table (**Static & Dynamic**) view
 - Contains best routes (active routes):
 - Connected, static, and dynamic routes
 - Doesn't contain:
 - Inactive, standby, and policy routes



The routing monitor widget on the dashboard page enables you to view the routing table and policy route table entries. The routing table contains *the best routes* (or active routes) of the following types:

- Static: manual routes that are configured by the administrator.
- Connected: automatic routes added by FortiOS after an interface is assigned an IP address. A connected route references the interface IP address subnet.
- Dynamic: routes learned using a dynamic routing protocol such as BGP or OSPF. FortiGate installs these routes automatically in the routing table and indicates the dynamic routing protocol used.

To view the routing table entries, select **Static & Dynamic**, as shown on this slide. However, keep in mind that the routing table doesn't contain the following routes:

- Inactive routes: These are static and connected routes with interfaces that are administratively down or with links that are down. Static routes are marked inactive when the link health monitor detects that their gateway is dead.
- Standby routes: These are active routes that are removed from the routing table because they are duplicate and have higher distances. Some examples of standby routes are:
 - A second static default route with a higher distance than another static default route
 - A dynamic route such as BGP or OSPF, to the same destination as another static route. However, the dynamic route is not displayed in the routing table because the static route has a lower distance.
- Policy routes: These include regular policy routes, ISDB routes, and SD-WAN rules. Policy routes are viewed in a separate table—the policy route table. You will learn more about policy routes in another lesson.

Route Attributes

- Each route in the routing table has the following attributes:

- Network
- Gateway IP
- Interfaces
- Distance
- Metric
- Priority

Dashboard > Network > Static & Dynamic Routing

Network	Gateway IP	Interfaces	Distance	Type	Metric
0.0.0.0/0	10.0.11.254	port2	10	Static	0
10.0.3.0/24	10.0.11.254	port2	120	RIP	2
10.0.4.0/24	10.0.11.254	port2	110	OSPF	2
10.0.5.0/24	10.0.11.254	port2	20	BGP	0
10.0.11.0/24	0.0.0.0	port2	0	Connected	0
192.168.0.0/16	0.0.0.0	port1	0	Connected	0

Enable the Metric column (disabled by default)

Best Fit Columns

Reset Table

Export

Select Columns

- ✓ Network
- ✓ Gateway IP
- ✓ Interfaces
- ✓ Distance
- ✓ Type
- ✓ Metric
- Priority
- Origin
- Tunnel ID
- Up Since
- VRF

Apply Cancel

get router info routing-table all

Codes: K - kernel, C - connected, S - static

...output omitted...

Routing table for VRF=0

S* 0.0.0.0/0 [10/0] via 10.0.11.254, port2, [1/0]
 R 10.0.3.0/24 [120/2] via 10.0.11.254, port2, 00:00:03, [1/0]
 O 10.0.4.0/24 [110/2] via 10.0.11.254, port2, 00:11:29, [1/0]
 B 10.0.5.0/24 [20/0] via 10.0.11.254 (recursive is directly connected, port2), 01:04:00, [1/0]
 C 10.0.11.0/24 is directly connected, port2
 C 192.168.0.0/24 is directly connected, port1

Display routing table entries on the CLI

Each of the routes listed in the routing table includes several attributes with associated values.

The **Network** column lists the destination IP address and subnet mask to match. The **Interfaces** column lists the interface to use to deliver the packet.

The **Distance**, **Metric**, and **Priority** attributes are used by FortiGate to make various route selection decisions. You will learn about each of these in this lesson.

This slide also shows the command you can run to display the routing table on the FortiGate CLI. The `get router info routing-table all` command displays the same route entries as the routing monitor widget on the FortiGate GUI.

Distance

- First tiebreaker for duplicate routes (best route selection)
 - The lower the distance, the higher the preference
 - Set by the administrator (except connected routes)
- Best route selection
 - Route with lowest distance is installed in the RIB
 - Standby routes (higher distance) are not installed in the RIB
 - They are installed in the routing table database
- Avoid multiple equal-distance duplicate routes but different protocol
 - FortiGate keeps the route that was learned last

Distance, or administrative distance, is the first tiebreaker that routers use to determine the best route for a particular destination. If there are two or more routes to the same destination (duplicate routes), the lowest-distance route is considered the best route and, as a result, is installed in the routing table. Other lower-distance routes to the same destination are standby routes and, as a result, are not installed in the routing table. Instead, they are installed in the routing table database.

Distance (Contd)

- Default distance per route type:

Connected	Static (SD-WAN zone)	Static (DHCP)	Static (Manual)	Static (IKE)	EBGP	OSPF	IS-IS	RIP	IBGP
0	1	5	10	15	20	110	115	120	200

Dashboard > Network > Static & Dynamic Routing

Network ↕	Gateway IP ↕	Interfac... ↕	Distance ↕	Type ↕	Metric ↕
0.0.0.0/0	10.200.1.254	port1	10	Static	0
10.0.1.0/24	0.0.0.0	port3	0	Connected	0
10.0.3.0/24	10.0.1.200	port3	200	BGP	0
10.0.4.0/24	10.0.1.200	port3	110	OSPF	2
10.0.5.0/24	10.0.1.200	port3	120	RIP	2
10.200.1.0/24	0.0.0.0	port1	0	Connected	0
10.200.2.0/24	0.0.0.0	port2	0	Connected	0
172.16.100.0/24	0.0.0.0	port8	0	Connected	0

You can set the distance for all route types except connected and IS-IS routes—both are hardcoded and their distance value cannot change. This slide shows the default values per type of route.

In case FortiGate learns two equal-distance routes to the same destination but that are sourced from different protocols, then FortiGate installs in the routing table the route that was learned *last*. For example, if you set the distance of BGP routes to 110, and there is another OSPF route to the same destination using the default administrative distance (110), then FortiGate keeps whichever route was learned last in the routing table. Because this behavior can lead to different results based on the timing of events, then it's not recommended to configure different-protocol routes with the same distance.

Metric

- Tiebreaker for same-protocol duplicate dynamic routes
 - The lower the metric, the higher the preference
- Best route is installed in the routing table and other duplicate routes in the routing table database
- The calculation method differs among routing protocols

Dashboard > Network > Static & Dynamic Routing

Network ↕	Gateway IP ↕	Interfac... ↕	Distance ↕	Type ↕	Metric ↕
0.0.0.0/0	10.200.1.254	 port1	10	Static	0
10.0.1.0/24	0.0.0.0	 port3	0	Connected	0
10.0.3.0/24	10.0.1.200	 port3	200	BGP	0
10.0.4.0/24	10.0.1.200	 port3	110	OSPF	2
10.0.5.0/24	10.0.1.200	 port3	120	RIP	2
10.200.1.0/24	0.0.0.0	 port1	0	Connected	0
10.200.2.0/24	0.0.0.0	 port2	0	Connected	0
172.16.100.0/24	0.0.0.0	 port8	0	Connected	0

When a dynamic route protocol learns two or more routes to the same destination, it uses the metric as a tiebreaker to identify the best route. The lower the metric, the higher the preference. The dynamic routing protocol then installs the best route in the routing table and the higher-metric routes in the routing table database. Note that the metric is used as tiebreaker for same-protocol dynamic routes, and *not* between different-protocol dynamic routes.

The metric calculation differs among routing protocols, and the details are not covered in this course. For example, RIP uses the hop count, which is the number of routers the packet must pass through to reach the destination. OSPF uses cost, which is determined by the link bandwidth.

Priority

- Tiebreaker for static routes with same distance
 - All static routes with equal distance are installed in the routing table, even if they have different priorities
 - Routes with lower priorities are preferred routes
 - Default value: 1
- Best route is used during route lookup

Dashboard > Network > Static & Dynamic

Network	Gateway IP	Distance	Type	Metric	Priority
0.0.0.0/0	10.1.5.254	10	Static	0	1
0.0.0.0/0	10.1.0.254	10	Static	0	10
10.1.0.0/24	0.0.0.0	0	Connected	0	0
10.1.4.0/24	10.1.0.100	110	OSPF	2	1
10.1.5.0/24	0.0.0.0	0	Connected	0	0
100.0.0.0/8	10.1.5.254	120	RIP	2	1
100.64.1.0/24	10.1.0.254	20	BGP	0	1
100.75.5.1/32	10.1.5.254	120	RIP	2	1

Network > Static Routes

Edit Static Route

Destination **Subnet** Internet Service
0.0.0.0/0.0.0.0

Gateway Address
10.1.0.254

Interface
port1

Administrative Distance **10**

Comments
Write a comment... 0/255

Status **Enabled** Disabled

Advanced Options

Priority **10**

Priority set to 10
(default value is 1)

When there are two or more duplicate static routes that have the same distance, FortiGate installs all of them in the routing table.

The priority setting enables administrators to break the tie among *static* routes. During the route lookup process, FortiGate selects the best route—that is, the static route with the lowest priority among all the equal-distance, duplicate static routes. The lower the priority value, the higher the preference.

The priority attribute applies to all routes except connected routes and is set to 1 by default.

For dynamic routes, you can change the priority of BGP routes only. The priority of other dynamic routes is hardcoded to 1. In dynamic routes, the priority value is useful for advanced routing deployments involving SD-WAN and multiple virtual routing and forwarding (VRF) IDs. A detailed explanation of how the priority attribute is beneficial for such cases is outside the scope of this course.

For static routes, you can configure the priority settings in the **Advanced Options** section, as shown on this slide.

To view the priority in the routing monitor widget, you must enable the **Priority** column (disabled by default). You can also view the priority in the routing table on the FortiGate CLI.

Routing Table—CLI

```
# get router info routing-table all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
V - BGP VPNv4
* - candidate default

Routing table for VRF=0
S*  0.0.0.0/0 [10/0] via 10.200.1.254, port1, [1/0]
C    10.0.1.0/24 is directly connected, port3
C    10.200.1.0/24 is directly connected, port1
C    10.200.2.0/24 is directly connected, port2
C    172.16.100.0/24 is directly connected, port8
```

Priority/Weight

Source

Distance/Metric

The CLI command shown on this slide displays all entries in the routing table. The routing table displays the routes that make it the best active routes to a destination.

The left-most column indicates the route source. Route attributes are shown inside square brackets. The first number, in the first pair of attributes, is distance, which applies to both dynamic and static routes. The second number is metric, which applies to dynamic routes only.

Static routes and dynamic routes also have priority and weight attributes, which are shown as the last pair of attributes for the respective route. In the case of dynamic routes, the weight is always zero.

This command doesn't show standby or inactive routes, which are present in the routing table database only. For example, when two static routes to the same destination subnet have different distances, the one with the lower distance is installed in the routing table, and the one with the higher distance in the routing table database.

GUI Route Lookup Tool

- Look up route by:
 - Destination address (required)
 - Destination port, source address, source port, protocol, and source interface (optional)
- If all criteria are provided:
 - FortiGate checks both routing table and policy route table entries
 - Otherwise, FortiGate checks routing table entries only
- Matching route is highlighted

Dashboard > Network > Static & Dynamic Routing

Route Lookup Create Address Edit Search

Network	Gateway IP	Interfaces
0.0.0.0/0	10.200.1.254	port1
10.0.1.0/24	0.0.0.0	port3
10.200.1.0/24	0.0.0.0	port1
10.200.2.0/24	0.0.0.0	port2
172.16.100.0/24	0.0.0.0	port8

Route Lookup

Destination: 8.8.8.8

Destination port: 1-65535

Source: IP or FQDN

Source port: 1-65535

Protocol: TCP

Source Interface:

You are redirected to the policy page if you enter all attributes

Matching route

Route Lookup Create Address Edit Search

Network	Gateway IP	Interfaces	Distance	Type
0.0.0.0/0	10.200.1.254	port1	10	Static
10.0.1.0/24	0.0.0.0	port3	0	Connected
10.200.1.0/24	0.0.0.0	port1	0	Connected

You can perform a route lookup on the routing monitor widget by clicking **Route Lookup**. Then, you must indicate at least the destination address to look up for, and optionally, the destination port, source address, source port, protocol, and source interface.

The way the route lookup works is as follows:

- If you don't provide all lookup criteria, FortiGate considers only the routing table entries. FortiGate then highlights the matching route, if any.
- If you provide all lookup criteria, FortiGate considers both routing table and policy route table entries. If the lookup matches a policy route, the GUI redirects you to the policy route page, and then highlights the corresponding matching policy route.

The example on this slide shows a route lookup tool for 8.8.8.8 and TCP as destination address and protocol, respectively. Because the administrator doesn't provide all criteria, FortiGate considers the routing table entries only. Then, the route lookup highlights the static default route as the matching route.

Reverse Path Forwarding

- IP anti-spoofing protection
- Source IP is checked for a return path
- RPF check is only carried out on:
 - The first packet in the session, not on a reply
- Two modes:
 - Feasible path (default; formerly loose)
 - Return path doesn't have to be the best route
 - Strict
 - Return path must be the best route
- If RPF check fails, debug flow shows:
 - `reverse path check fail, drop`

- Set RPF mode (default = disable):

```
config system settings
  set strict-src-check [disable | enable]
end
```

Strict mode

- Disable RPF (default = enable):

```
config system interface
  edit <interface>
    set src-check disable
  next
end
```

The RPF check is a mechanism that protects FortiGate and your network from IP spoofing attacks by checking for a return path to the source in the routing table.

The premise behind the RPF check is that if FortiGate receives a packet on an interface, and FortiGate doesn't have a route to the packet source address through the incoming interface, then the source address of the packet could have been forged, or the packet was routed incorrectly. In either case, you want to drop that unexpected packet, so it doesn't enter your network.

FortiGate performs an RPF check only on the first packet of a new session. That is, after the first packet passes the RPF check and FortiGate accepts the session, FortiGate doesn't perform any additional RPF checks on that session.

There are two RPF check modes:

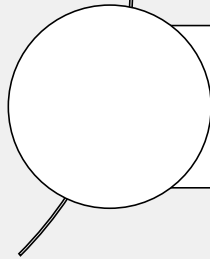
- Feasible path: Formerly known as loose, it's the default mode. In this mode, FortiGate verifies that the routing table contains a route that matches the source address of the packet and the incoming interface. The matching route doesn't have to be the best route in the routing table for that source address. It just has to match the source address and the incoming interface of the packet.
- Strict: In this mode, FortiGate also verifies that the matching route is the best route in the routing table. That is, if the routing table contains a matching route for the source address and incoming interface, but there is a better route for the source address through another interface, then, the RPF check fails.

This slide also shows how to change the RPF check mode on the FortiGate CLI, as well as how to disable the RPF check on the interface level.

Lesson Progress



Routing on FortiGate



ECMP Routing

Good job! You now understand routing on FortiGate.

Now, you will learn about equal cost multipath (ECMP) routing.

ECMP Routing

Objectives

- Implement route redundancy and load balancing

After completing this lesson, you should be able to achieve the objective shown on this slide.

By demonstrating competence in ECMP routing, you will be able to implement routing load balancing.

ECMP

- Same-protocol routes with equal:
 - Destination subnet
 - Distance
 - Metric
 - Priority
- ECMP routes are installed in the RIB
 - Traffic is load balanced among routes

So far, you've learned about the different route attributes that FortiGate looks at to identify the best route to a destination.

But what happens when two or more routes of the same type have the same destination, distance, metric, and priority? These routes are called equal cost multipath (ECMP) routes, and FortiGate installs all of them in the routing table. FortiGate also load balances the traffic among the ECMP routes.

ECMP (Contd)

Dashboard > Network > Static & Dynamic

Network ↕	Gateway IP ↕	Interfaces ↕	Distance ↕	Type ↕	Metric ↕	Priority ↕
0.0.0.0/0	10.200.1.254	port1	10	Static	0	5
0.0.0.0/0	10.200.2.254	port2	10	Static	0	5
10.0.1.0/24	0.0.0.0	port3	0	Connected	0	0
10.0.2.0/24	0.0.0.0	port4	0	Connected	0	0
10.0.3.0/24	10.0.1.200	port3	200	BGP	0	1
10.0.3.0/24	10.0.2.200	port4	200	BGP	0	1
10.0.4.0/24	10.0.1.200	port3	110	OSPF	2	1
10.0.4.0/24	10.0.2.200	port4	110	OSPF	2	1
10.200.1.0/24	0.0.0.0	port1	0	Connected	0	0
10.200.2.0/24	0.0.0.0	port2	0	Connected	0	0

Two ECMP static routes

Two ECMP BGP routes

Two ECMP OSPF routes

```
# get router info routing-table all
...output omitted...
Routing table for VRF=0
S* 0.0.0.0/0 [10/0] via 10.200.1.254, port1, [5/0]
   [10/0] via 10.200.2.254, port2, [5/0]
C 10.0.1.0/24 is directly connected, port3
C 10.0.2.0/24 is directly connected, port4
B 10.0.3.0/24 [200/0] via 10.0.1.200, (recursive is directly connected, port3), 00:07:04, [1/0]
   [200/0] via 10.0.2.200, (recursive is directly connected, port4), 00:07:04, [1/0]
O 10.0.4.0/24 [110/2] via 10.0.1.200, port3, 00:15:12, [1/0]
   [110/2] via 10.0.2.200, port4, 00:15:12, [1/0]
C 10.200.1.0/24 is directly connected, port1
C 10.200.2.0/24 is directly connected, port2
```

The example on this slide shows two ECMP static routes, two ECMP BGP routes, and two ECMP OSPF routes. For each ECMP group, the destination subnet, distance, metric, and priority are the same.

The result is that FortiGate installs both routes of each ECMP group in the routing table. This lesson, however, focuses on ECMP static routes only.

ECMP Load Balancing Algorithms

- Source IP (default)
 - Sessions sourced from the same address use the same route
- Source-destination IP
 - Sessions with the same source *and* destination address pair use the same route
- Weighted
 - Applies to static routes only
 - Sessions are distributed based on route, or interface weights
 - The higher the weight, the more sessions are routed through the selected route
- Usage (spillover)
 - One route is used until the bandwidth threshold is reached, then the next route is used

ECMP can load balance sessions using one of the following four algorithms:

- Source IP: This is the default algorithm. FortiGate uses the same ECMP route to route sessions sourced from the same address.
- Source-destination IP: FortiGate uses the same ECMP route to route sessions with the same source-destination IP address pair.
- Weighted: Applies to static routes only. FortiGate load balances sessions based on the route weight or the respective interface weight. The higher the weight, the more sessions FortiGate routes through the selected route.
- Usage (spillover): FortiGate sends sessions to the interface of the first ECMP route until the bandwidth of the interface reaches the configured spillover limit. After the spillover limit is reached, FortiGate uses the interface of the next ECMP route.

Configuring ECMP

- If SD-WAN is disabled, the ECMP algorithm is set on the CLI:

```
config system settings
    set v4-ecmp-mode [source-ip-based | weight-based | usage-based | source-dest-ip-based]
end
```

- Configure weight values on the CLI on the interface level (left) and route level (right):

```
config system interface
    edit <interface name>
        set weight <0-255>
    next
end
```

Interface weight has precedence over the weight of a static route using the interface

```
config router static
    edit <id>
        set weight <0-255>
    next
end
```

- Configure spillover thresholds on the CLI (kbps):

```
config system interface
    edit <interface name>
        set spillover-threshold <0-16776000>
        set ingress-spillover-threshold <0-16776000>
    next
end
```

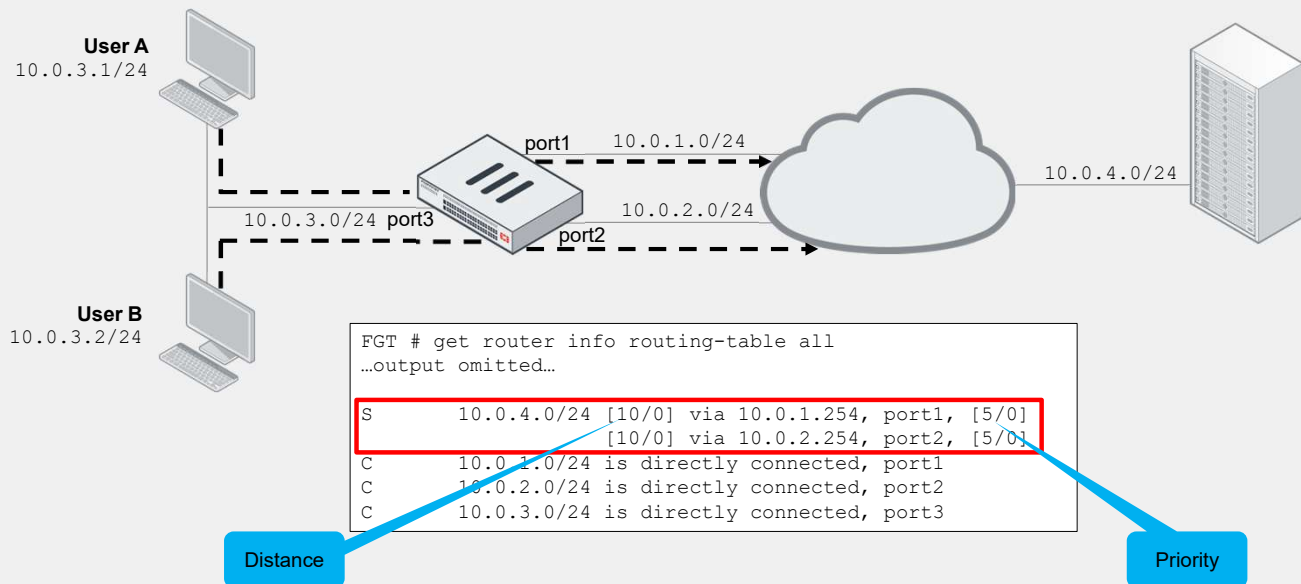
If SD-WAN is disabled, you can change the ECMP load balancing algorithm on the FortiGate CLI using the commands shown on this slide.

When SD-WAN is enabled, FortiOS hides the `v4-ecmp-mode` setting and replaces it with the `load-balance-mode` setting under `config system sdwan`. That is, when you enable SD-WAN, you control the ECMP algorithm with the `load-balance-mode` setting.

For spillover to work, you must also configure the egress and ingress spillover thresholds, as shown on this slide. If you disable SD-WAN, FortiGate selects the lowest numbered interface first. The thresholds are set to 0 by default, which disables spillover check.

For a weighted algorithm, you must configure the weights at the interface level or route level, as shown on this slide. If two or more routes are added to the routing table, and you set `v4-ecmp-mode` to `weight-based`, FortiGate routes sessions based on the weight value of each route.

ECMP Example



In the scenario shown on this slide, FortiGate has ECMP routes for the 10.0.4.0/24 subnet on port1 and port2. Using the default ECMP algorithm (source IP based), FortiGate may use any of the two routes to route traffic from user A and user B.

In the example shown on this slide, FortiGate selects the route over port1 for user A, and the route over port2 for user B. FortiGate continues to use the same selected routes for the same traffic. If the route over port1 is removed from the routing table, FortiGate automatically starts to forward the traffic sourced from both users and destined to 10.0.4.0/24 through port2.

ECMP enables you to use multiple paths for the same destination, as well as provide built-in failover. Usually, you want to use ECMP for mission-critical services that require high availability. Another reason to use ECMP is for bandwidth aggregation. That is, you can leverage the bandwidth of multiple links by load balancing sessions across them.

While ECMP enables you to leverage multiple WAN links on FortiGate, you may want to use SD-WAN because of the additional benefits.

Default ECMP Algorithm vs. SD-WAN ECMP Algorithm

ECMP (<code>v4-ecmp-mode</code>)	SD-WAN (<code>load-balance-mode</code>)
Both control ECMP algorithms	
Not available when SD-WAN is enabled	Not available when SD-WAN is disabled
Doesn't support volume algorithm	Support volume algorithm
Uses the weight defined in the static route	Uses the SD-WAN member weight
Uses the interface spillover thresholds	Uses the SD-WAN member spillover thresholds

- Volume algorithm:
 - FortiGate tracks the cumulative number of bytes of the member
 - The higher the member weight, the higher the target volume, the more traffic is sent to it

When you enable SD-WAN, FortiOS hides the `v4-ecmp-mode` setting and replaces it with the `load-balance-mode` setting under `config system sdwan`. That is, after you enable SD-WAN, you now control the ECMP algorithm with the `load-balance-mode` setting.

There are some differences between the two settings. The main difference is that `load-balance-mode` supports the volume algorithm, and `v4-ecmp-mode` does not. In addition, the related settings such as weight and spillover thresholds are configured differently. That is, when you enable SD-WAN, the weight and spillover thresholds are defined on the SD-WAN member configuration. When you disable SD-WAN, the weight and spillover thresholds are defined on the static route and interface settings, respectively.

When you set the ECMP algorithm to volume—this is when SD-WAN is enabled, FortiGate load balances sessions across members based on the measured interface volume and the member weight. That is, the volume algorithm instructs FortiGate to track the cumulative number of bytes of each member and to distribute sessions based on the weight. The higher the weight, the higher the target volume of the interface and, as a result, the more traffic FortiGate sends to it.

Knowledge Check

1. The priority attribute is a tiebreaker for which type of routes?
☒ A. Static
☐ B. Connected
2. Which attribute does FortiGate use to determine the *best* route for same-protocol, duplicate dynamic routes?
☐ A. Priority
☒ B. Metric
3. What is the default ECMP algorithm on FortiGate?
☐ A. Weighted
☒ B. Source IP

Lesson Progress



Routing on FortiGate



ECMP Routing

Congratulations! You have completed this lesson.

Now, you will review the objectives covered in this lesson.

Review

- ✓ Configure static routing
- ✓ Interpret the routing table on FortiGate
- ✓ Implement route redundancy and load balancing

This slide shows the objectives that you covered in this lesson.

By mastering the objectives covered in this lesson, you learned how to configure, monitor, and load balance routes on FortiGate.